

EdgeTrust-VANET

Edge-Assisted Trust-Based Secure Communication in VANETs

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- Vehicular Ad Hoc Networks (VANETs) enable wireless communication among vehicles and Road Side Units (RSUs).
- Designed primarily for safety-critical and traffic-efficiency applications.
- Vehicles exchange messages related to accidents, congestion, hazards, and road conditions.
- Communication must be reliable, low-latency, and secure.

- Traditional VANET security focuses on encryption and authentication (PKI, certificates).
- Authentication verifies identity but cannot guarantee honest behavior.
- Insider attacks occur when authenticated vehicles send false or misleading data.
- This creates a need for behavior-aware trust and validation mechanisms.

Motivation

- False safety alerts may trigger unnecessary braking, rerouting, congestion, or panic situations, directly impacting road safety.
- Insider attacks can bypass traditional identity-based defenses despite valid authentication.
- Centralized validation systems introduce latency, single points of failure, and scalability limitations in high-density traffic scenarios.
- Sole reliance on RSUs may lead to computational overhead and performance bottlenecks under heavy traffic conditions.
- Distributed edge-assisted decision-making enables faster, localized, and scalable trust validation.
- Integrating Trust Management, Edge Intelligence, and Machine Learning enhances robustness, reliability, and adaptability in dynamic vehicular environments.

- A Lightweight Machine Learning Model to Detect VANET Attacks.
[Connected Vehicles Security \(2025\)](#)
- Trust-based schemes evaluate credibility using reputation/history.
[Vehicular Communication Systems – MDPI \(2025\)](#)
- Machine learning used for anomaly/misbehavior detection.
[IEEE VANET Security Study \(2024\)](#)
- Edge computing proposed for latency-sensitive vehicular systems.
- However, integration among these components remains limited.

Detailed Survey Document:

[View Literature Survey](#)

- No unified framework combining Trust, Edge Computing, and ML.
- Most systems rely only on RSUs without considering scalability.
- Centralized ML solutions increase latency.
- Limited solutions for insider attack detection.
- Trust decisions are not enforced through adaptive forwarding.
- Hazard alert systems lack integrated trust validation.

Edge-Assisted Trust-Based Secure Communication in VANETs

- Algorithm to calculate trust scores for vehicles based on behavior, message consistency, and historical data.
- Multi-layer edge processing using OBUs, RSUs, and fog layer with SDN-based control for real-time hazard validation and scalable workload distribution.
- Simulation or prototype demonstrating secure message exchange using trust-based filtering.
- Analysis using metrics such as latency, packet delivery ratio, detection accuracy, and false alert reduction.

Novelty of Proposed Work

- Unified Trust + Edge + ML framework.
- Hierarchical multi-layer architecture (OBU-RSU-Fog).
- RSUs as intelligent edge validation nodes.
- Blockchain-based secure trust synchronization.
- SDN-based adaptive forwarding control.

Existing Algorithms Overview

Security Algorithms	Trust Algorithms	ML Algorithms
PKI Authentication	Trust-based Routing	SVM
Certificate Revocation (CRL)	Reputation Scoring	Random Forest
Position Verification	Credibility Aggregation	Isolation Forest
Group Signature Schemes	Bayesian Trust Model	K-Nearest Neighbors
Timestamp Validation	Weighted Trust Models	Ensemble Learning

Datasets Planned

- VeReMi (VANET Misbehavior Dataset) [View Dataset](#)
- VANET-MaliciousNode Dataset [View Dataset](#)
- Data sets with traces for VANET [View Dataset](#)
- VANET Threat Dataset [View Dataset](#)
- Pothole Detection Dataset [View Dataset](#)

Implementation Overview

- The system is implemented as a modular pipeline combining:
 - Dataset processing
 - Trust score computation
 - Machine learning classification
 - Web-based dashboard
- Designed for real-time evaluation at Road Side Units (RSUs)
- Ensures low latency and scalable decision-making

- Dataset: **vanet_malicious_nodes.csv**
- Total records: **5000 vehicles**
- Malicious nodes: **25.2%**
- Attack types:
 - Blackhole
 - Sybil
 - DoS
 - Injection

Features Used

- Mobility Features:
 - Position (X, Y)
 - Speed, Direction
 - Acceleration
- Network Features:
 - Packets Sent / Received
 - Packet Drop Ratio
 - Latency, Signal Strength
- Trust Features:
 - Direct Trust
 - Neighbor Trust
 - Historical Trust

- Script: **analyze_vanet_nodes.py**
- Removes data leakage features (attack labels)
- Trains multiple ML models:
 - Random Forest
 - Gradient Boosting
 - Extra Trees
 - SVM, KNN, Logistic Regression
- Dataset split into training and testing sets

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1 Score (Primary metric)

Best Model: Random Forest

- Accuracy: 95.6%
- F1 Score: 95.71%
- High robustness to noisy data

Trust Score Computation

- Script: **trust_score.py**
- Calculates trust score between 0 and 1
- Based on:
 - Message consistency
 - Physical plausibility
 - Historical behavior

- Trust + ML output combined
- Final decision:
 - **ACCEPT** → Normal behavior
 - **WARN** → Suspicious activity
 - **BLOCK** → Malicious node

- File: **dashboard/app.py**
- Framework: Python Flask
- Provides APIs:
 - `/api/evaluate`
 - `/api/vanet_models`
- Connects ML model and trust module to UI

- Vehicle sends data to RSU
- Trust score is computed
- Features passed to ML model
- Model predicts:
 - Normal / Malicious
- Final decision generated:
 - ACCEPT / WARN / BLOCK
- Output displayed on dashboard

- SUMO – Traffic mobility simulation.
- OMNeT++ / Veins – VANET network simulation with SDN support.
- Python + Scikit-learn – Trust scoring and ML models.
- Docker Environment – Fog layer deployment simulation.

Justification for OBU and Fog Layer

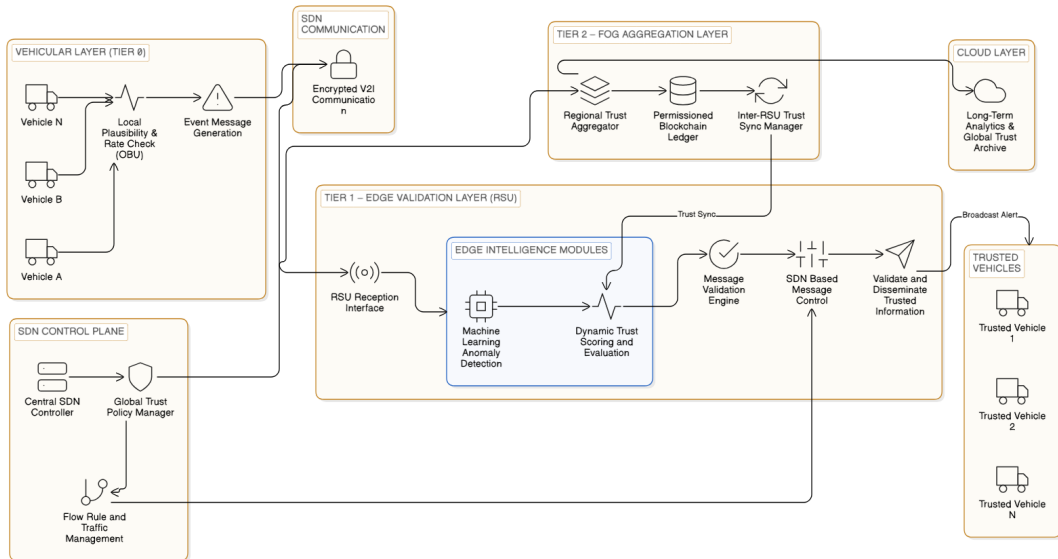
OBU-Level Support

- OBUs can initiate and verify safety events before forwarding to RSUs, reducing unnecessary traffic load.
B-SAFE (2024): Blockchain-Enabled Security for Vehicular Fog.Link
- Hardware-assisted OBU authentication and plausibility verification improve trust reliability.
Blockchain Trust Mgmt in VANETs (2025). Link
- Distributed OBU/RSU validation mechanisms enhance detection of false position and spoofing attacks.
False Position Attacks Detection (2025). Link

Fog Layer Support

- Fog nodes enable regional blockchain synchronization between RSUs for scalable trust management.
B-SAFE (2024). Link
- Consensus-based RSU coordination improves trust consistency across infrastructure.
Blockchain Trust Mgmt in VANETs (2025). Link
- Fog-based ML ensembles enhance anomaly detection while reducing RSU overload.
False Position Attacks Detection (2025). Link

System Architecture



- **Phase 1: Trust Modeling & Algorithm Design** Design trust scoring and anomaly detection models.
- **Phase 2: Multi-Layer Edge Implementation** Implement OBU-level local validation, RSU-based trust processing, SDN control, and fog-based trust synchronization.
- **Phase 3: Performance Evaluation & Validation** Analyze system performance under various attack scenarios.

Conclusion

- This work proposes a hierarchical trust-adaptive framework for secure VANET communication.
- The architecture combines multi-layer edge validation with OBUs, RSUs and fog-based aggregation.
- Lightweight ML and SDN-based forwarding are considered for adaptive trust enforcement.
- Permissioned blockchain is explored for secure trust synchronization across infrastructure.

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Thank You